# Statistics 280 Winter 2025 Midterm 2

| Name (Please print legibly) | UMID Number |
|---|---|
| | |

- Please answer all questions in the space provided and indicate where in the exam you have written additional work.
- **Solutions without justification will be marked as wrong for all questions except Question 5 and other questions asking for R code. Show all work. A calculator command does not count as justification**.
- Question 7 is an extra credit question.
- Express answers as fractions or as decimals rounded to 2 decimal places.
- The last page is a table of values from the normal distribution.

## Academic Integrity

Please read and sign the following statement regarding academic integrity to indicate that you have read, understand, and comply with the statement.

I have never given nor received unauthorized aid on this examination, nor have I concealed any violations of the academic integrity policy as stated by the College of Literature, Science, and the Arts. I have not done anything that may be understood as any action or attempted action that may result in creating an unfair academic advantage for oneself or an unfair academic advantage or disadvantage for any other member of members of the academic community. I understand that if I as suspected of academic misconduct, the incident may be reported to the Office of the Assistant Dean.

Signature: _____

1.(30 points) Suppose that you draw cards from a standard 52 card deck **without** replacement until you draw a spade.

a) What is the chance that you draw exactly 1 card?

b) What is the chance that you draw exactly 2 cards?

c) What is the chance that you draw exactly 3 cards?

d) What is the chance that you draw more than 5 cards?

e) What is the smallest number of cards where the chance of drawing that many cards or fewer is at least 90%?

2. (30 points) Suppose that 49% of undergraduate students at the University of Michigan are out of state and that 21% of undergraduate students are first year students.

a) What is the maximum percentage of undergraduate students who are out of state or first year? What would be the relationship between being out of state and a first-year student in this case?

b) Suppose that 12% of undergraduates are both out of state and first year students. What percentage of undergraduates are out of state or first year students?

c) Would you conclude that being out of state and being a first-year student are independent in part b)? Why or why not?

d) Suppose that there is some characteristic where 49% of undergraduates at the University of Michigan have that characteristic or are out of state students. What can you say about that characteristic?

3. (30 points) Suppose that you are studying 3 species of butterfly in a particular region: A, B, and C. Based on previous studies of the region, you estimate that 30% of butterflies in the region are species A, 50% species B, and 20% species C. You have a butterfly specimen from the region with a wing pattern that is found in 60% of species A butterflies, 20% of species B butterflies, and 90% of species C butterflies. What is the chance that your butterfly specimen is from species A, from species B, and from species C based on the wing pattern?

4. (40 points) Suppose that you are the director of a non-profit organization and have a pool of volunteers consisting of 5 men and 10 women. You have 5 events and you need 2 volunteers to staff each event. You also have a rule that each volunteer can only volunteer for one event.

a) If you randomly choose the 10 volunteers, then the chance that exactly 3 of the volunteers are men is _____.

b) Write the R command to calculate the chance that at least 5 of the volunteers are women. Include all inputs in your R command.

_____.

c) Suppose that you relax your rule and allow a volunteer to staff multiple events. If you randomly choose the 10 volunteers, then the chance that exactly 3 of the volunteers are men is _____. Count a given volunteer as multiple volunteers if they volunteer for multiple events.

d) Write the R command to calculate the chance that at most 4 of the volunteers are men. Include all inputs.

_____.

e) The chance that more than 2 of the volunteers in part c are men is _____.

f) Write the R command to calculate the chance in part c) with all inputs.

_____.

5. (30 points) a) Write an R function called *cointoss* that takes the probability *p* as input and outputs *1* (heads) with probability *p* and *0* (tails) with probability *1-p*.

b) The function below simulates drawing from a *0-1* box without replacement until drawing a *1*. It returns the number of draws required to draw a *1*. Fill in the blanks in the function

```
# n0 – number of 0's, n1 – number of 1's
draw1=function(n0, n1) {
        # p is the chance of drawing a 1
        p = _____
        # You want to keep drawing until there are only 1's left in the box
        for (i in 1: _____) {
                draw = cointoss(p)
                # You drew a 1
                if (_____)  {

                        return(_____)
                }
                else {
                        # You drew a 0. Remove the 0 from the box for the next draw.
                        # How many 0's have you drawn total to get to this step?
                        p = _____
                }
        }
}
```

c) Fill in the blanks in the code below. This code uses the function in 5b) to estimate the expected value of the number of cards needed to draw a spade in problem 1) based on 1,000,000 simulated values.

```
# Preallocate vector to store the number of cards drawn for each simulation.
cards_drawn = numeric(1,000,000) # Vector with 1,000,000 empty elements
# Simulate the experiment
for (_____) {
  cards_drawn[__] = draw1(___ , __)
}
# Calculate the estimated expected value.
expected_value = _____
```

6. (40 points) The Boston Celtics from the National Basketball League won 64 out of 82 games in the 2024 regular season. Suppose that we are trying to predict how many games that they will win in 2025. Assume that the Celtics have a 64/82 = 78% chance of winning each game in 2025 and that the 82 games are independent.

a) The expected value and standard error for the number of games that the Celtics win in 2025 are _____ and _____, respectively.

b) Write an R command with all inputs to calculate the exact chance that the Celtics win 60 games or more in 2025 based on the assumptions above.

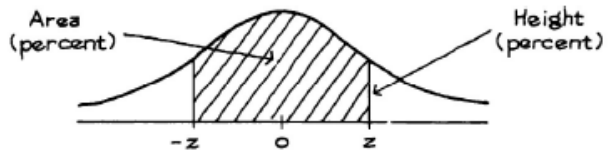
_____.

c) The chance that the Celtics win 60 games or more is approximately _____, assuming that the number of wins is approximately normally distributed. You will want to use 59.5 wins in your calculations due to the continuity correction.

d) The Celtics record at this point in the 2025 season is 49-19. What is the chance that the Celtics would have 49 or fewer wins after 68 games if the assumptions in part a) are true? Are the assumptions in part a) realistic given this result? Use the normal approximation with continuity correction.

7. (15 points extra credit) Suppose that $X$ and $Y$ are random variables with possible values of $0$ and $1$. Show that $E(X + Y) = E(X) + E(Y)$.

# Tables




A NORMAL TABLE

| *z* | *Height* | *Area* | *z* | *Height* | *Area* | *z* | *Height* | *Area* |
|---|---|---|---|---|---|---|---|---|
| 0.00 | 39.89 | 0 | 1.50 | 12.95 | 86.64 | 3.00 | 0.443 | 99.730 |
| 0.05 | 39.84 | 3.99 | 1.55 | 12.00 | 87.89 | 3.05 | 0.381 | 99.771 |
| 0.10 | 39.69 | 7.97 | 1.60 | 11.09 | 89.04 | 3.10 | 0.327 | 99.806 |
| 0.15 | 39.45 | 11.92 | 1.65 | 10.23 | 90.11 | 3.15 | 0.279 | 99.837 |
| 0.20 | 39.10 | 15.85 | 1.70 | 9.40 | 91.09 | 3.20 | 0.238 | 99.863 |
| 0.25 | 38.67 | 19.74 | 1.75 | 8.63 | 91.99 | 3.25 | 0.203 | 99.885 |
| 0.30 | 38.14 | 23.58 | 1.80 | 7.90 | 92.81 | 3.30 | 0.172 | 99.903 |
| 0.35 | 37.52 | 27.37 | 1.85 | 7.21 | 93.57 | 3.35 | 0.146 | 99.919 |
| 0.40 | 36.83 | 31.08 | 1.90 | 6.56 | 94.26 | 3.40 | 0.123 | 99.933 |
| 0.45 | 36.05 | 34.73 | 1.95 | 5.96 | 94.88 | 3.45 | 0.104 | 99.944 |
| 0.50 | 35.21 | 38.29 | 2.00 | 5.40 | 95.45 | 3.50 | 0.087 | 99.953 |
| 0.55 | 34.29 | 41.77 | 2.05 | 4.88 | 95.96 | 3.55 | 0.073 | 99.961 |
| 0.60 | 33.32 | 45.15 | 2.10 | 4.40 | 96.43 | 3.60 | 0.061 | 99.968 |
| 0.65 | 32.30 | 48.43 | 2.15 | 3.96 | 96.84 | 3.65 | 0.051 | 99.974 |
| 0.70 | 31.23 | 51.61 | 2.20 | 3.55 | 97.22 | 3.70 | 0.042 | 99.978 |
| 0.75 | 30.11 | 54.67 | 2.25 | 3.17 | 97.56 | 3.75 | 0.035 | 99.982 |
| 0.80 | 28.97 | 57.63 | 2.30 | 2.83 | 97.86 | 3.80 | 0.029 | 99.986 |
| 0.85 | 27.80 | 60.47 | 2.35 | 2.52 | 98.12 | 3.85 | 0.024 | 99.988 |
| 0.90 | 26.61 | 63.19 | 2.40 | 2.24 | 98.36 | 3.90 | 0.020 | 99.990 |
| 0.95 | 25.41 | 65.79 | 2.45 | 1.98 | 98.57 | 3.95 | 0.016 | 99.992 |
| 1.00 | 24.20 | 68.27 | 2.50 | 1.75 | 98.76 | 4.00 | 0.013 | 99.9937 |
| 1.05 | 22.99 | 70.63 | 2.55 | 1.54 | 98.92 | 4.05 | 0.011 | 99.9949 |
| 1.10 | 21.79 | 72.87 | 2.60 | 1.36 | 99.07 | 4.10 | 0.009 | 99.9959 |
| 1.15 | 20.59 | 74.99 | 2.65 | 1.19 | 99.20 | 4.15 | 0.007 | 99.9967 |
| 1.20 | 19.42 | 76.99 | 2.70 | 1.04 | 99.31 | 4.20 | 0.006 | 99.9973 |
| 1.25 | 18.26 | 78.87 | 2.75 | 0.91 | 99.40 | 4.25 | 0.005 | 99.9979 |
| 1.30 | 17.14 | 80.64 | 2.80 | 0.79 | 99.49 | 4.30 | 0.004 | 99.9983 |
| 1.35 | 16.04 | 82.30 | 2.85 | 9.69 | 99.56 | 4.35 | 0.003 | 99.9986 |
| 1.40 | 14.97 | 83.85 | 2.90 | 0.60 | 99.63 | 4.40 | 0.002 | 99.9989 |
| 1.45 | 13.94 | 85.29 | 2.95 | 0.51 | 99.68 | 4.45 | 0.002 | 99.9991 |